MILITARY SPECIFICATION

INDICATOR COUPLER CU-395/ARN-21A

This specification has been approved
by the Bureau of Aeronautics, Department of the Navy

1. SCOPE AND CLASSIFICATION

1.1 Scope - The equipment covered by this specification shall be designed to couple the outputs of Receiver-Transmitter RT-220/ARN-21 to flight instruments for presentation of bearing and distance of an aircraft to a navigation beacon.

1.2 Classification - The equipment covered by this specification shall be of one type and shall consist of the following:

Units	Type Designation	Max. Allow. Size (Inches) L	W	H/D	Max. Allow. Weight Lbs.	Applicable Paragraphs
Indicator Coupler	CU-395/ARN-21A	8-5/8	7-3/8	4-5/16	10.4	3.3.1
Mounting						
Mounting Base	MT-1530/ARN-21A	7-5/16	8-7/8	1-5/8	0.6	3.3.2

1.3 Auxiliary Equipment - The equipment shall operate with the following auxiliary equipments; however, these equipments are not to be supplied as part of this equipment:

(1) Receiver-Transmitter	RT-220/ARN-21
(2) Course Indicator	ID-249A/ARN, ID-351A/ARN, ID-357/ARN, or ID-387/ARN
(3) Course Indicator	ID-250/ARN
(4) Distance Indicator	ID-388/APN
(5) Navigational Computer	CP-( )/ARN-21

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

2.2. Specifications

Military

MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-T-5422	Testing; Environmental, Aircraft and Electronic Equipment
MIL-W-5088	Wiring, Aircraft, Installation of

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MIL-E-7894	Electric Power, Aircraft, Characteristics of.
MIL-I-6181	Interference Limits, Tests and Design Requirements Aircraft Electrical Equipment.
JAN-P-658	Packaging and Packing of Electrical Equipment and Spare Parts (Electronic, Electric and Electro-Mechanical).
JAN-T-152	Treatment, Moisture-and-Fungus-Resistant, of Communications, Electronic and Associated Electrical Equipment: General Process for.
MIL-N-18307	Nomenclature and Nameplate for Electronic and Associated Equipment.
MIL-T-18303	Test Procedures, Preparation of.

Standards

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-200	Electron Tubes

2.3 Availability of Documents

(1) When requesting specifications, publications and drawings refer to both title and number. All requests should be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of this specification and applicable specifications and publications may be obtained upon application to the Commanding Officer, Naval Air Station, Jonnsville, Pennsylvania, Attn: *Technical Records Division.*

(3) Copies of the U. S. Navy Reliable Tube List may be obtained upon application to Bureau of Aeronautics, Navy Department, Washington 25, D. C., Attn: Electronics Division.

2.4 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the Procuring Agency.

(3) Referenced Specifications- Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials- In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following factors shall govern:

(1) Parts and materials as approved by specification MIL-E-5400 shall be given first consideration.

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(2) When the contractors can demonstrate that the use of standard AN or JAN parts or materials will not fulfill the design objectives because of size, weight, performance or other reasons, materials and parts shall be used which most nearly meet or exceed the requirements of the respective standard specification.

(3) When a non-standard part is used, the meeting of general material requirements such as non-inflammability, fungus resistance, non-toxicity, etc., and, the meeting of environmental conditions as required by specification MIL-E-5400 or MIL-T-5422 shall be of prime consideration.

(4) In the selection of parts, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event the entire assembly is built to be expended, the assembly shall be constructed so that the unit as a whole will pass material, electrical and environmental tests, without consideration to the quality of the detail parts. Any assembly built to be expended shall be first approved by the Bureau of Aeronautics.

3.1.1 Non-Standard Part Approval and Material Approval - Approval for the use of non-standard parts and materials shall be obtained as outlined in specification MIL-E-5400 except as follows:

(1) Mechanical parts are expected to meet the material requirements of MIL-E-5400, and shall be submitted as non-standard parts only when the material requirements are not met.

(2) Non-standard parts, as defined in specification MIL-E-5400 of an electrical nature must be approved before the preproduction model of the equipment is delivered to the Navy. The request for approval must be forwarded by the contractor before the part is purchased or fabricated. In addition to or in lieu of the non-standard part data sheet the contractor is requested to submit itemized lists of the parts, their sources, and the contractor's drawing or purchase specification covering the parts.

3.1.2 Design Objectives - Minimum size and weight, simplicity of operation and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives in the design of the equipment specified herein. The use of materials, parts and processes other than those required by Specification MIL-E-5400 shall be investigated and when it appears that a substantial reduction in size and weight or improvement in simplicity of design, performance or reliability can be realized by their use, a request for approval of a non-standard part shall be submitted to the Bureau of Aeronautics for consideration. Each request should be accompanied by complete supporting information.

3.1.3 Plug-in Assemblies - The use of easily disconnected, plug-in assemblies of one or more related circuits shall be investigated with a view towards reduction of size and weight, simplification of operation and maintenance or improvement in performance.

3.1.4 Choice of Tube Types - Electronic tubes used in the equipment shall be chosen from the following listings in the sequence noted:

(1) U. S. Navy Reliable Tube List

(2) MIL-STD-200

The use of any tube not on these acceptable listings must be approved by the Bureau of Aeronautics; however, before consideration for approval, the contractor must submit the following information:

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(a) Quantitative data on the electrical advantages and mechanical adequacy of the proposed tube over the most nearly equivalent type on the acceptable listings.

(b) Test data and performance limits of the tube which must be maintained to insure satisfactory and reliable performance in the equipment.

(c) Information as to whether the tube manufacturer has received Military or Navy qualification approval for the particular tube.

3.1.5 Workmanship - Workmanship, fabrication and manufacturing processes shall conform to the requirements of MIL-E-5400.

3.2 General Design Requirements

3.2.1 General - This equipment shall fulfill all applicable design and performance requirements of Specification MIL-E-5400 unless otherwise specified herein.

3.2.2 Service Conditions - The equipment shall operate satisfactorily under any of the service conditions or reasonable combinations of service conditions as specified in Specification MIL-E-5400.

3.2.2.1 Vibration - The equipment shall be required to meet the vibration requirement of 5 - 500 cps with vibration isolators in place, instead of vibration isolators removed, as required by MIL-E-5400.

3.2.3 Total Weight - The total weight of the equipment including mounting base, excluding cables and connectors, shall not exceed 11 pounds.

3.2.4 Operational Stability - The equipment shall operate with optimum performance for 500 hours, continuously or intermittently, without necessity for readjustment of any controls which are non-accessible during flight.

3.2.5 Operating Life

3.2.5.1 Reliable Operating Life - The equipment shall have a reliable operating life of at least 500 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life should be specified by the manufacturer.

3.2.5.2 Total Operating Life - The equipment shall have a minimum total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.6 Interconnecting Cabling - This equipment shall be capable of satisfactory operation using external wiring in accordance with Specification MIL-W-5088. The external wiring shall be unshielded except that a minimum number of the individual wires may be shielded, when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished.

3.2.7 Cables and Connectors - The equipment shall be designed for the use of cables and connectors in accordance with the requirements of Specification MIL-E-5400. However, cables and that portion of the connectors which are to be attached to the cables shall not be supplied as part of the equipment.

3.2.8 Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-E-7894 and shall give specified performance from the following power sources with characteristics as defined in MIL-E-7894 having limits specified herein. The power required shall not exceed the specified amounts.

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(1) AC Power (Single Phase)

(a) 115 Volt Constant Frequency

Operating Voltage Limits	110 to 120 volts
Operating Frequency Limits	380 to 420 cps
Reduced Performance Voltage Limits	102 to 110 and 120 to 124 volts
Reduced Performance Frequency Limits	360 to 380 and 420 to 440 cps
Power (1Ø)	40 V.A.

(2) DC Power 28 Volts System

Operating Voltage Limits	24 to 29 volts
Power	0.1 amps

3.2.9 Under Voltage Protection - The equipment shall not be damaged by voltages below the minimum specified and shall automatically resume normal operation when the voltage returns within limits

3.2.10 Moisture and Fungus Resistance Treatment - Equipments shall be fungus proofed by selection of parts and materials as non-nutrient for fungus or the parts and materials shall be treated prior to their use in the equipment so that over-all spraying of the equipment is not necessary. Over-all spraying of the equipment in accordance with JAN-T-152 shall be made in the event selection of parts and materials described above is not possible.

3.2.11 Nomenclature and Name Plates - Nomenclature assignment and name plate approval for equipment identification shall be in accordance with the requirements of Specification MIL-N-18307.

3.2.12 Interchangeability - The interchangeability requirements as specified in Specification MIL-E-5400 shall be extended to require interchangeability of units and component parts of this equipment with the latest model of any previously produced equipments. Such interchangeability shall be measured against a model, manufacturing drawings, or other technical information as provided by the Bureau of Aeronautics for that purpose. In the event that requirements of MIL-E-5400 conflict with the requirements for interchangeability the interchangeability requirements shall govern.

3.2.13 Provisions for Maintenance and Field Testing - Provisions for maintenance and field testing shall meet the requirements of Specification MIL-E-5400.

3.2.14 Radio Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.15 Performance - With adjustment and testing in accordance with the approved Test Procedures for Indicator-Coupler CU-395/ARN-21A and under the test conditions specified in Section 4.2 of MIL-E-5400, the accuracy of the presented information shall fall within the following limits:

3.2.15.1 Indicated Bearing Accuracy - Bearing information accuracy shall fall within the following limits:

(1) Fine Bearing Dial - With the FINE bearing dial set to 0.0 ±0.2° the bearing measuring resolver shall be zeroed as a phase shifter at 135 cps.

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(2) Coarse Bearing Dial - When the FINE bearing dial is set to the zero corresponding to zero degrees of bearing the COARSE bearing dial shall read zero.

3.2.15.2 Relative and Magnetic Bearing Output Accuracy - The bearing input to the Indicator Coupler shall be represented by a test signal applied to the bearing measuring resolver. Bearing output information shall not deviate from the bearing input by more than 1°, with the differential synchro transmitter zeroed at 180°.

3.2.15.3 Fundamental Synchro Resolver Accuracy - The output of the Fundamental Synchro Resolver shall be within ±0.75° of bearing input represented by a position test signal applied to the bearing measuring resolver. The Fundamental Synchro Resolver shall be zeroed at 300° and be readily adjustable over the range ±3°.

3.2.15.4 Harmonic Bearing Resolver - The output of the Harmonic Bearing Resolver (if supplied) shall be within 0.1° (one speed) of the bearing input as represented by a position test signal applied to the bearing measuring resolver.

3.2.15.5 Indicated Distance Accuracy - Distance information as indicated on the COARSE and FINE Distance Indicator dials shall fall within the following limits:

(1) With the FINE distance dial set to 1±0.05 miles the distance measuring resolver shall be zeroed as a phase shifter at 4044 cps

(2) Coarse Distance Dial - When the FINE distance dial is set to the zero corresponding to zero miles the COARSE distance dial shall read zero.

3.2.15.6 Distance Take-Off Accuracy - The output resistance of the unloaded take-off potentiometers shall be proportional to the distance measured to within ±0.3% of maximum resistance from zero to 194 nautical miles with minimum output resistance occurring at -2.0±0.05 miles. This test shall be referenced to a position test signal applied to the distance measuring resolver.

3.2.15.7 Distance Synchro Transmitter Output Accuracy - The distance input to the Indicator Coupler shall be represented by a position test signal applied to the distance measuring resolver. Synchro transmitter output accuracy shall be checked by use of a test circuit connected to the UNITS distance synchro transmitter. The difference between input and output information shall not exceed ±.05 miles.

3.2.15.8 Sensitivity - The sensitivity of both the bearing and the distance mechanisms shall be such that with the application of 18 volts, 400 cps, to the fixed phase of either motor, and the rate generator phase 1 excited with 18 volts, 400 cps, the application of 2 volts, 400 cps leading or lagging 90°, to the control phase of that same motor will cause the motor to start with the indicator dials in any position and continue to rotate smoothly.

3.2.15.9 Indicator Speed - With 18 volts, 400 cps applied to both phases of either motor, and the rate generator phase 1 excited with 18 volts, 400 cps, and with the voltage across the control phase either leading or lagging the voltage across the fixed phase by 90° that motor shall drive its corresponding indicator either backward or forward at the following rate:

(1) Bearing Mechanism 27.5° per second ±15%.

(2) Distance Mechanism 11.5 miles per second ±15%.

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3.2.15.10 Course Deviation Sensitivity - With three standard deviation indicator loads (1000 ohms each), the course deviation output circuit shall be adjustable to a maximum sensitivity of at least 5° course error equal to a full scale deviation indication of 150 microamperes per indicator (10 degrees total course width pin to pin).

3.2.15.11 To-From Circuit - The To-From circuit shall be capable of supplying the ambiguity indicator on the ID-249/ARN with direct current of either polarity and a magnitude of 250 ±250 microamperes. Polarity of current to be determined by relative phase of 400° cps signal input to To-From circuit.

3.3 Detail Design Requirements

3.3.1 Indicator Coupler CU-395/ARN-21A - The unit shall be designed to meet the following requirements:

3.3.1.1 Function - Indicator Coupler CU-395/ARN-21A when properly interconnected to Radio Receiver-Transmitter RT-220/ARN-21 and components supplying the required signal and control voltages shall perform the following functions:

(1) Supply signal information to Course Indicator ID-250/ARN in such a manner that it will continuously indicate the magnetic bearing of a navigation beacon from the receiver-transmitter.

(2) With compass input from a low impedance source, provide an output to indicate by means of a pointer on Course Indicator, ID-250/ARN the relative bearing of a navigation beacon from the receiver-transmitter.

(3) Provide a control circuit for remotely switching between (1) and (2) above.

(4) Provide a fundamental synchro resolver output which will supply a navigation computer or Course Indicator ID-249A/ARN with bearing information.

(5) Provide phase detectors operating off the output of the Course Selector resolver situated in the Course Indicator ID-249A/ARN, phase detector outputs to be utilized to show magnitude and direction of angular deviation from selected course on the vertical pointer of ID-249A/ARN, and to resolve the course ambiguity by operating the To-From Indicator of the ID-249A/ARN. The deviation output shall be designed to operate into an impedance equal to three standard deviation indicator loads in parallel.

(6) Provide for addition of a harmonic synchro resolver to operate at nine times the rate of the fundamental synchro resolver.

(7) Automatically and continuously supply required information concerning the distance of the receiver-transmitter from a navigation beacon to Distance Indicator ID-388/AP N. One, two, or three Distance Indicators may be employed.

(8) Provide a means for obtaining a voltage proportional to distance for use in a navigational computer.

3.3.1.2 Form Factor and Mounting

3.3.1.2.1 Housing - The equipment shall be completely housed in a pressurized case approximately 7-3/8" wide by 4-5/16" deep by 8-5/8" long. All electrical connection to the unit shall be through two connectors mounted upon one 7-1/8 x 4-5/16 area. An outline drawing of the housing is shown in Figure 1.

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3.3.1.2.2 Case - The case shall be made of a light weight alloy and finished on all surfaces in dull black wrinkle finish. Suitable transparent windows shall be located upon one surface of the case as to permit a clear view of the bearing and distance indicator dials. The case shall be secured to the frame of the equipment by means of screws located around the edge of the 7-3/8" x 4-5/16" surface containing all electrical connections. It shall be designed in such a manner as to support the enclosed equipment.

3.3.1.2.3 Mounting - The equipment shall be mounted by means of four mounting pads to the mounting base MT-1530/ARN-21A.

3.3.1.3 Contents of Unit - The equipment shall be designed with the following basic component parts and sub-assemblies:

3.3.1.3.1 Motor Rate Generators - A two phase induction motor-rate generator with a motor rating of 26 volts, 400 cycles per second, per phase and a generator rating of 0.3 volt alternating current output at a speed of 1000 rpm shall be designed to drive the bearing indicator dials, bearing measuring potentiometer, differential transmitter and resolvers through associated gearing and supply an alternating voltage proportional to speed of rotation. A similar motor-rate generator shall drive the distance indicator dials, distance measuring resolver, distance take-off transmitters and distance measuring and take-off potentiometers through associated gearing. The motor portion of the motor-rate generator shall be designed to operate with 26 volts ±10%, 380 to 420 cycles per second with a reversible phase from 90° lagging to 90° leading applied to the control phase of the motor. The rate generator shall be designed to operate with 18 volts ±10%, 380 to 420 cps continuously applied to phase one. Each motor-rate generator shall be electrically and mechanically interchangeable with Eclipse-Pioneer Type FV-102-0 or an approved equivalent.

3.3.1.3.2 Bearing Measuring Potentiometer - A precision sinusoidal potentiometer rated 150 volts, 16,000 ohms ±10%, angular accuracy ±.6°, amplitude accuracy ±.8% with 360° of rotation with approximately 85% of the total resistance within the brush circle shall be designed to determine the phase difference between two 15 cps signals for measuring bearing in conjunction with the bearing measuring resolver. The bearing measuring potentiometer shall be designed to maintain specified performance with 15 volts, 15 cps applied to the card continuously. The bearing measuring potentiometer shall be electrically and mechanically interchangeable with Gamewell Type 68815-1 or an approved equivalent.

3.3.1.3.3 Bearing Measuring Resolver - A precision resolver shall be designed to determine the phase difference between two 135 cps signals for measuring bearing in conjunction with the bearing measuring potentiometer. The bearing measuring resolver shall be designed to maintain specified performance with 0.7 volt, 135 cps applied to the rotor continuously. The resolver shall be electrically and mechanically interchangeable with Bureau of Ordnance Type 11R16N2.

3.3.1.3.4 Differential Synchro Transmitter - A synchro transmitter of the differential control type rated at 11.8 volts per phase, 400 cycles per second shall be designed to produce a signal with a space phase equal to the space phase difference between the compass heading signal and the RT-220/ARN-21 bearing signal. The synchro transmitter shall be electrically and mechanically interchangeable with Bureau of Ordnance Type 26V-11CDX-4A.

3.3.1.3.5 Fundamental Synchro Resolver - A synchro resolver rated at 26 volts, 400 cycles per second with a two-phase rotor and a two-phase stator shall be designed to produce an output directly related to the bearing dials. The synchro resolver shall be supplied with 26.5 volts, 380 to 420 cps for continuous operation. The synchro resolver shall be electrically and mechanically interchangeable with Bureau of Ordnance Type 11R2N4.

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3.3.1.3.6 Distance Measuring Potentiometer - A precision potentiometer cup rated at 3 watts nominal and 2 watts at 85°C, 20,000 ohms ±1%, linearity 0.2% for 350° with 360° of rotation and a dead angle of 5.5° or less shall be designed to measure distance in conjunction with the distance measuring resolver. The distance measuring potentiometer shall be designed to maintain specified performance with 90 volts ±10% direct current continuously applied. The potentiometer assembly shall be electrically and mechanically interchangeable with Gamewell Type 6845-1 or an approved equivalent.

3.3.1.3.7 Distance Take-Off Potentiometers - Two potentiometer cups with ratings of 3 watts nominal and 2 watts at 85°C, 2000 ohms ±1%, linarity 0.15% for 350°, with 360° of rotation and a dead angle of 5.5° or less shall be designed for use in computer circuits to obtain voltage proportional to distance. The distance take-off potentiometers shall be ganged with distance measuring potentiometer (3.3.1.3.6) and shall be designed to maintain specified performance with either an alternating or a direct current supply of 75 volts nominal and 63 volts at 85°C. The potentiometer assembly shall be electrically and mechanically interchangeable with Gamewell Type 68345-1 or an approved equivalent.

3.3.1.3.8 Distance Measuring Resolver - A precision resolver shall be designed for use in conjunction with the distance measuring potentiometer to measure distance. The resolver shall maintain specified performance with 4.5 volts, 4044 cycles, applied to the stator circuits or 9 volts, 4044 cycles applied to the rotor circuits. The resolver shall be electrically and mechanically interchangeable with Bureau of Ordnance Type 11R10N40.

3.3.1.3.9 Distance Synchro Transmitter - A synchro transmitter rated at 26 volts, 400 cycles input shall be designed to provide electrical distance output signals to the synchro torque receivers driving the units and tens dials of the ID-388/APN indicator. This synchro transmitter shall be electrically and mechanically interchangeable with the Bureau of Ordnance Type 26V-11TX-4A.

3.3.1.3.10 Hundreds Dial Switch - A switch shall be provided actuated by the distance gear train for the purpose of controlling the hundreds dial of the ID-388/APN. When the measured distance is between 0 and 99 miles, 400 cycle positioning voltages shall be supplied so as to cause a zero indication of the hundreds dial of the ID-388/APN. When the measured distance is between 100 and 195 miles the switch shall be actuated so as to cause an indication of one on the hundreds dial of the ID-388/APN.

3.3.1.3.11 Indicators - Distance and bearing indicators shall each consist of two rotating disks moving against fixed index lines. Each disk shall be approximately one inch in diameter, finished in dull black with contrasting numerals.

3.3.1.3.11.1 Bearing Indicator - The bearing indicator shall consist of two disks, the coarse bearing indicator dial and the fine bearing indicator dial. The coarse bearing indicator dial shall be divided into 9 equal arcs of 40° each and make one complete revolution for each revolution of the bearing measuring potentiometer. The fine bearing indicator dial shall be divided into 40 equal arcs of 9° each and make one revolution for each revolution of the bearing measuring resolver, (one dial division for each degree of indicated bearing).

3.3.1.3.11.2 Distance Indicator - The distance indicator shall consist of two disks, the coarse distance indicator dial and the fine distance indicator dial. The coarse distance indicator dial shall consist of two disks, the coarse distance indicator dial and the fine distance indicator dial. The coarse distance indicator dial shall be divided into 20 equal arcs of 18° each and shall make one complete revolution for each revolution of the distance measuring potentiometer. The fine distance indicator dial shall be divided into 10 equal arcs of 36° each and make 20 revolutions for each revolution of the distance measuring potentiometer, (one dial division for each one mile of distance).

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3.3.1.3.12 Power Transformer - The Indicator Coupler CU-395/ARN-21A shall include a self-contained power supply. The power transformer, in addition to supplying voltage for DC requirements, shall be capable of furnishing 26.5 volts at 400 cycles per second for transmitter and fundamental bearing resolver synchro excitation; 12 volts and 4.9 volts at 400 cycles per second for supplying input voltage to the differential synchro transmitter in the magnetic bearing indicator and the hundreds dial synchro receiver of ID-388/ARN; 5 volts (center tapped and isolated) at 400 cycles per second for phase detector biasing, and 6.3 volts at 400 cycles per second for filament excitation.

3.3.1.3.13 Relative-Magnetic Relay - A 28 volts DC relay shall be designed to select the proper circuits to cause Course Indicator ID-250/ARN to depict either relative or magnetic bearing of the navigational beacon from the aircraft.

3.3.1.3.14 Connectors - Two 47 pin connectors shall be designed to provide all external connections to the equipment. The connectors shall mate with plugs conforming to Air-Force-Navy Aeronautical Drawing AND 10435.

3.3.1.1.15 Connector Arrangement - The arrangement of pin functions of the external connectors shall be in accordance with the following:

Input Connector J-201

Pin	Function
A	No connection.
B	No connection.
C	No connection
D	Bearing measuring resolver rotor black.
E	No connection.
F	No connection.
G	Bearing measuring resolver stator black.
H	No connection.
I	Bearing measuring resolver stator red.
J	Bearing measuring resolver stator yellow.
K	No connection.
L	No connection.
M	No connection.
N	Bearing measuring resolver rotor red.
O	Bearing measuring resolver stator blue.
P	Distance measuring potentiometer series resistance.
R	Distance measuring resolver stator yellow.
S	Compass Input.
T	Distance measuring resolver stator black.
U	Distance measuring resolver stator red.
V	Distance measuring resolver stator blue.
W	No connection.
X	Compass Input.
Y	Distance motor-generator common.
Z	Distance measuring resolver rotor red.
a	Distance measuring resolver rotor yellow.
b	Bearing measuring potentiometer input "F".
c	Bearing motor - Phase #2.
d	Distance motor - Phase #1.
e	Distance measuring resolver rotor common.
f	Bearing measuring potentiometer output "D".
g	Bearing measuring potentiometer output "G".
h	Distance motor - Phase #2.
j	Distance rate generator - Phase #1.
k	Bearing motor generator common.

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m	Distance measuring potentiometer input "T_1".
n	Bearing measuring potentiometer input "E".
p	Bearing motor - Phase #1.
r	Distance rate generator - Phase #2 Gnd.
s	Distance measuring potentiometer output "W".
t	Distance rate generator output.
u	Bearing rate generator output.
v	Distance measuring potentiometer input "T_2".
w	Bearing rate generator - Phase #2 Gnd.
x	Search/track signal.
y	Bearing rate generator - Phase #1.
z	27.5 volts DC input.

Output Connector J-202

Pin	Function
A	Fundamental resolver stator blue.
B	Fundamental resolver stator yellow.
C	Deviation sensitivity control jumper.
D	Fundamental resolver stator black.
E	Units distance transmitter stator "S_3".
F	No. 1 distance take-off potentiometer "T_2".
G	Fundamental resolver stator red.
H	No. 2 distance take-off potentiometer "W".
I	Fundamental resolver rotor red.
J	Fundamental resolver rotor yellow.
K	Relative/magnetic control.
L	Deviation sensitivity control jumper.
M	Relative/magnetic output.
N	115 volts 400 cps input.
O	Fundamental resolver rotor common black.
P	Tens distance transmitter stator "S_1".
R	No. 1 distance take-off potentiometer "T_1".
S	To-From input.
T	Relative/magnetic output.
U	Tens distance transmitter stator "S_3"
V	Differential stator "S_3".
W	No. 2 distance take-off potentiometer "T_2".
X	Differential stator "S_1".
Y	No. 2 distance take-off potentiometer "T_1".
Z	Deviation sensitivity control arm.
a	27.5 volts DC output.
b	Indicator common.
c	Hundreds synchro stator "S_1".
d	Hundreds synchro stator "S_2".
e	6 volts, 400 cps.
f	Deviation input.
g	No connection.
h	Ground.
j	Search/track signal.
k	No. 1 distance take-off potentiometer "W".
m	No connection.
n	No connection.
p	Units distance transmitter stator "S_1".

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Output Connector J-202

Pin	Function
r	26.0 volts, 400 cps fundamental resolver excitation.
s	No connection.
t	Deviation output.
u	To-From output.
v	No connection.
w	Hundreds synchro excitation.
x	No connection.
y	B+ output at 20 ma.
z	Hundreds synchro excitation.

3.3.2 Mounting Base MT-1530/ARN-21A

3.3.2.1 Function - Mounting base MT-1530/ARN-21A shall provide a means of mounting Indicator Coupler CU-395/ARN-21A to the airframe or radio equipment rack.

3.3.2.2 Form Factor and Mounting - Mounting Base MT-1530/ARN-21A shall be designed as a formed sheet metal plate to mount four all metal vibration isolators. Maximum dimensions shall be 8-7/8 inches wide, 7-5/16 inches long and 1-5/8 inches high, as shown in Figure 2.

4 QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests - The testing and inspection of aircraft electronic equipment shall be classified as follows:

(1) Preproduction Tests - Preproduction tests are those tests conducted on samples which are representative of the production equipment to be supplied under the contract, for the purpose of determining that the equipment meets all the requirements of the applicable specifications. Preproduction tests shall consist of the following two tests:

(a) Contractor Demonstration Tests - Contractor Demonstration Tests are accomplished under the responsibility of the contractor on a sample, representative of the production equipment, to demonstrate that the equipment complies with all the applicable requirements.

(b) Service Approval Tests - Service Approval Tests are accomplished under the responsibility of the procuring service on a sample which is representative of the production equipment, to determine that the equipment meets all the requirements of this specification. These tests may be accomplished by having the procuring service supervise the Contractor Demonstration Tests.

(2) Inspection Tests - Inspection Tests are those tests accomplished on equipment submitted for acceptance under the contract. Inspection Tests shall consist of the following:

(a) Individual Tests - Individual Tests are those tests conducted on each equipment submitted for acceptance under the contract.

(b) Sampling Tests - Sampling Tests are those tests accomplished on a quantity of equipment as approved by the Bureau of Aeronautics and which have previously passed the Individual Tests.

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(c) Special Tests - Special Tests are those tests accomplished on a quantity of equipment as approved by the Bureau of Aeronautics and which have previously passed the Sampling Tests.

(3) Life Tests - Life Tests shall be conducted by the contractor at his plant for the purpose of determining the reliability of the equipment as well as its capability to meet the requirements of operating life and stability. These tests shall be started as soon as practicable on an early production equipment.

4.2 Preproduction Tests

4.2.1 Contractor Demonstration Tests - The contractor shall conduct tests on one or more sample equipments as necessary to determine that the design of the equipment as proposed by the contractor will meet the requirements of all applicable specifications. Contractor Demonstration Tests shall be conducted in accordance with the approved preproduction test procedure. The data obtained by the contractor in conducting these tests shall be included with the design data submitted with the Service Approval Test sample. The Inspector of Naval Material and the Bureau of Aeronautics shall be advised when tests are to be conducted so that the Bureau of Aeronautics may designate a representative to witness or supervise the test when so desired. When tests are supervised the Bureau of Aeronautics may decide to omit certain portions of the Service Approval Tests which will duplicate the Contractor Demonstration Tests; however, in conducting Service Approval Tests the Bureau of Aeronautics may elect to repeat any test previously conducted by the contractor, when deemed necessary. Contractors not having laboratory facilities to satisfactorily conduct all tests shall obtain either the services of a commercial testing laboratory or written approval from the Bureau of Aeronautics to omit that portion of the tests which cannot be conducted.

4.2.2 Service Approval Tests - The Service Approval sample shall consist of one equipment representative of the production equipment to be supplied under the contract. This equipment shall be delivered by the contractor to the destination specified by the Bureau of Aeronautics.

4.2.2.1 Accessory Materials Required - In addition to the complete equipment submitted, the contractor shall submit the accessory material and design data as specified in Specification MIL-E-5400.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the model meets all the requirements of this specification and the contract. Preproduction tests shall include testing in accordance with Specification MIL-T-5422 and MIL-I-6181. The design and test data supplied by the contractor shall indicate the equipments physical and electrical characteristics and establish that the equipment meets all the requirements of this specification. The test data shall include data obtained by the contractor in testing to MIL-T-5422.

4.2.4 Preproduction Sample Approval - Approval of the preproduction sample shall be by the Bureau of Aeronautics upon satisfactory completion of all tests. No additional equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of any equipments prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample will be returned to the contractor for his use in the fabrication and testing of the equipment to be submitted for acceptance under the contract. The preproduction sample shall not be considered as one of the equipments under the contract, however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.3 Inspection Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the required tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having laboratory testing facilities satisfactory to the Government Inspector shall engage the services of a commercial testing laboratory acceptable to the Bureau of Aeronautics. The contractor shall furnish test reports showing quantitative results for all inspection tests required by this specification, and signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guarantee of the accept-

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ance of the finished product. Inspection Tests shall consist of the following:

(1) Individual Tests - Individual Tests shall be conducted on each equipment submitted for acceptance under the contract.

(2) Sampling Tests - Sampling Tests shall be conducted on a quantity of equipment as specified in paragraph 4.3.2.1. Equipments for sampling tests shall be selected by the Government Inspector after the equipment has passed the individual tests and have been packaged and packed ready for shipment.

(3) Special Tests - Special Tests shall be conducted on a quantity of equipment as specified in paragraph 4.3.3.1. Equipments for Special Tests shall be selected by the Government Inspector after the equipment has passed the Sampling Tests.

4.3.1 Individual Tests - Each equipment shall be examined carefully to determine compliance with the requirements of this specification with respect to material and workmanship. In addition, the equipment shall be given an operational test, with sufficient data recorded to assure satisfactory operation.

4.3.1.1 Scope of Test - As a minimum each equipment shall be tested to determine compliance with the following:

	Individual Tests	Applicable Paragraph
I.	Tests conducted under standard laboratory conditions and nominal input voltages and frequency.	
1.	Check material and workmanship	
2.	Indicated Bearing Accuracy	3.2.15.1
3.	Relative and Magnetic Bearing Output Accuracy	3.2.15.2
4.	Fundamental Synchro Resolver Accuracy	3.2.15.3
5.	Harmonic Bearing Resolver Accuracy (If Supplied)	3.2.15.4
6.	Indicated Distance Accuracy	3.2.15.5
7.	Distance Take-Off Accuracy	3.2.15.6
8.	Distance Synchro Transmitter Output Accuracy	3.2.15.7
9.	Motor Sensitivity	3.2.15.8
10.	Indicator Speed	3.2.15.9
11.	Course Deviation Sensitivity	3.2.15.10
12.	To-From Circuit	3.2.15.11

4.3.2 Sampling Tests - Sampling Tests shall be conducted under the close supervision of the Government Inspector and shall include a performance check at ambient laboratory conditions. This Test may also include any of the other preproduction tests the Inspector deems necessary to determine conformance of the equipment with the requirements of this Specification. These tests shall include also a careful examination of the way the equipment is packaged and packed to determine conformance with the packaging and packing requirements of this specification or the contract. Sampling Tests shall be conducted on complete equipments. If individual units of the

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equipments are individually tested and shipped then the samples selected for Sampling Tests shall consist of all the individual units necessary to compose a complete operating equipment.

4.3.2.1 Sampling Test Schedule - Sampling Tests shall be performed on the following quantity of equipment unless otherwise stated in the contract or approved in writing by the procuring service.

Quantity of Equipment	Quantity of Equipment to be tested
First 10 equipments	1
Next 10 to 100	2
Next 100 to 200	1
Over 200	One for each additonal 200

4.3.2.2 Scope of Test - As a minimum, each equipment selected for sampling tests shall be tested to determine compliance with the following:

Sampling Tests		Applicable Paragraph
I.	Tests conducted under standard laboratory conditons and normal input voltages and frequency.	
	1. Check Material and Workmanship	
	2. Input Power	3.2.8
	3. Fundamental Synchro Resolver Accuracy	3.2.15.3
	4. Harmonic Bearing Resolver Accuracy	3.2.15.4
	5. Distance Synchro Transmitter Output Accuracy	3.2.15.7
	6. Course Deviation Sensitivity	3.2.15.10
	7. To-From Circuit	3.2.15.11
II.	Tests conducted as specified above using highest and lowest specified input voltages and frequencies.	

4.3.2.3 Equipment Failure - Should an equipment fail to pass any part of the Sampling Test, the following corrective action shall be taken by the contractor.

(1) Determine the cause of the failure.

(2) Determine if the failure is an isolated case or a design defect.

(3) If a design defect, suggest corrective action for BuAer approval.

(4) Include a test in the Individual Test to check all equipments for this requirement until reasonable assurance can be obtained that the defect has been satisfactorily corrected.

4.3.3 Special Test - Special Tests shall be conducted under the close supervision of the Government Inspector and shall include a performance check at the enviromental conditions specified below using the same test procedures as approved for the Preproduction Test. The Special Test may also include any other tests the Inspector deems necessary to determine conformance of the equipment with the contractual and specification requirements . Special Tests shall be conducted on complete equipments.

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4.3.3.1 Special Test Sampling Schedule - Special Tests shall be conducted on the first and every fifth equipment that successfully passes the Sampling Test.

4.3.3.2 Scope of Test - As a minimum each equipment selected for Special Test shall be tested to determine compliance with the following:

Special Test	Applicable Paragraph
I. Tests using procedures as approved for the preproduction test under the following environmental conditions:	
1. High and low temperature	3.2.2
2. Humidity	3.2.2

4.3.3.3 Equipment Failure - Should an equipment fail to pass any part of the Special Tests, the following corrective action shall be taken by the contractor:

(1) Determine the cause of the failure.

(2) Determine if the failure is an isolated case or design defect.

(3) If a design defect, suggest corrective action for BuAer approval.

4.4 Life Test - The equipment shall be subjected to a Life Test at the contractor's plant under the following simulated service conditions:

Temperature	-Normal room temperature
Altitude	-Normal (0 to 5000 ft.)
Humidity	-Room ambient
A.C. Voltage	-115 ±5 volts (at lowest applicable frequency)
D.C. Voltage	+27.5 ±2.5 volts

4.4.1 Test Data - The contractor shall keep a daily record as to the performance of the equipment, making particular note of any deficiencies or failures. In the event of parts failure during this test the defective parts shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all such failures, including all tube failures, throughout the test indicating (a) the part type number, (b) the circuit reference symbol number, (c) the part function, (d) the name of the manufacturer, (e) the nature of the failure, and (f) the number of hours which the part operated prior to the failure. A report shall be submitted to the Bureau of Aeronautics immediately in the event of a part failure or if no failure occurs at the completion of the test. In this report the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The Bureau of Aeronautics will review such proposals and determine whether they are acceptable.

4.4.2 Test Duration - The duration of the Life Test shall be 500 hours.

4.4.2.1 Test Periods - The life test may be run continuously or intermittently but no period of operation shall be less than six hours. At least one such period of operation shall be performed during each 24 working day. During each six or eight hour period the equipment shall be turned on and off several times and put through all phases of its performance.

4.4.2.2 Performance Check - At approximately eight hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the Bureau of Aeronautics.

4.4.3 Sampling Schedule - The equipment to be tested shall be representative of the current production in all significant aspects of design and manufacture. It

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shall be selected by the Government Inspector from equipments which have passed the Individual Test. The Life Test shall be performed on the following quantities of equipment unless otherwise stated in the contract or approved by the Bureau of Aeronautics.

Quantity of Item on order	Quantity of Items to be tested
1 - 100	1
100 - 500	2
Over 500	One for each additional 500

4.4.4 Reconditioning of Tested Samples - Equipment which has been subjected to the Life Test shall be reconditioned as follows:

(a) On completion of the Life Test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring agency.

(b) After reworking, the contractor may submit the equipment for acceptance under the contract.

4.5 Test Procedures - The procedures and methods for conducting Preproduction Tests, Inspection Tests and Life Tests shall be prepared by the contractor and sent to the Bureau of Aeronautics for approval. The right is reserved by the Bureau of Aeronautics or the Naval Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Use Specification MIL-T-18303 as a guide for preparation of Test procedures.

4.6 Presubmission Testing - Prior to the submission of any model for preproduction testing, the contractor shall furnish facilities for and shall conduct tests as required by paragraph 4.2.1. In addition, the proposed test procedure shall be submitted in sufficient time to permit its review by the Bureau of Aeronautics and all necessary revisions by the contractor prior to the start of any tests.

4.7 Rejection and Retest - The electronic equipment which has been rejected any be reworked or have parts replaced to correct the defects, and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring agency.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be packaged, packed and marked for shipment in accordance with the requirements of Specifications JAN-P-658 and MIL-STD-129.

6. NOTES

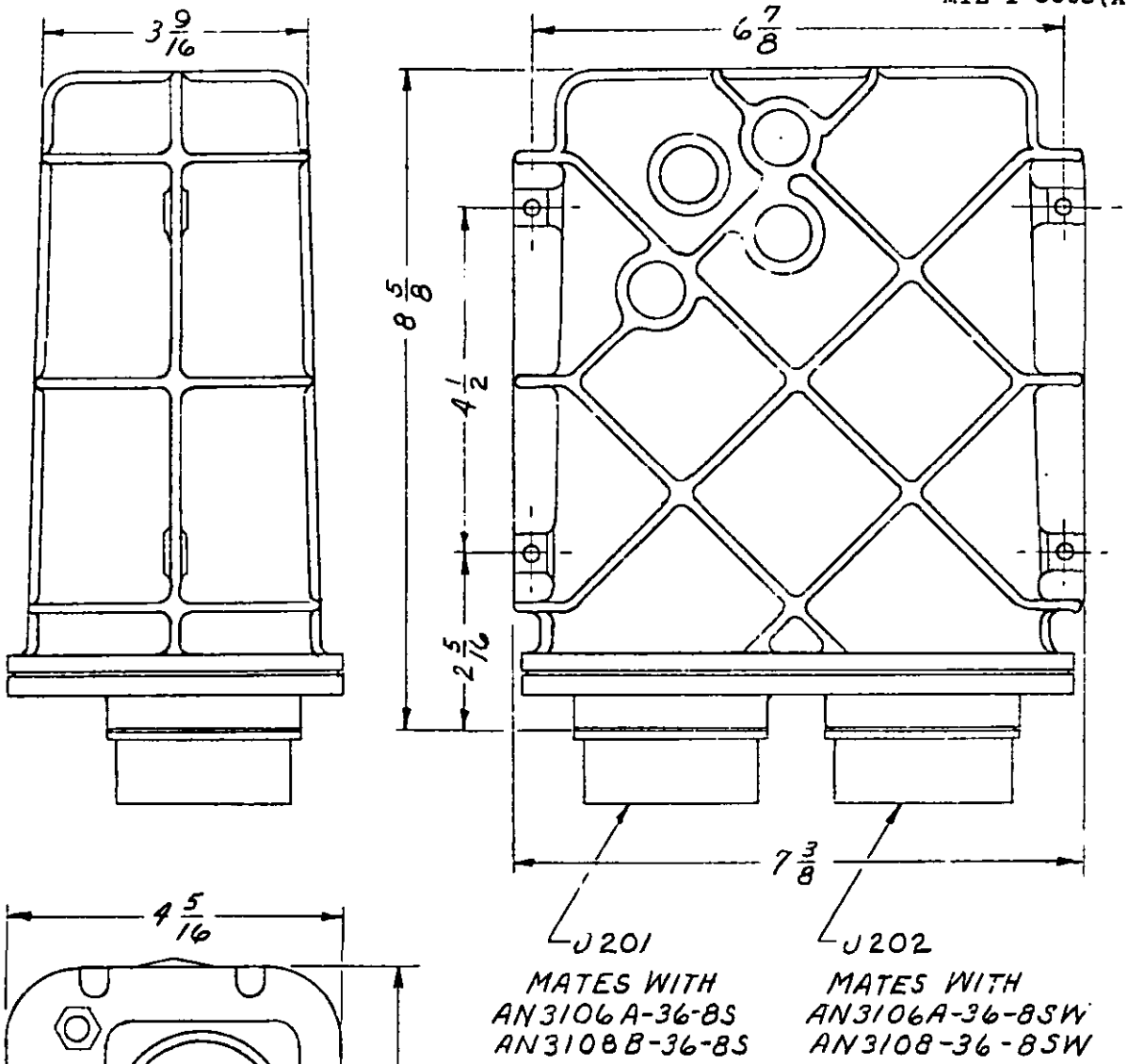
6.1 Use - The equipment shall be designed for use in Military aircraft to carry out specific military missions.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ±0.5 V DC (and 115 ±1 volt AC). These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

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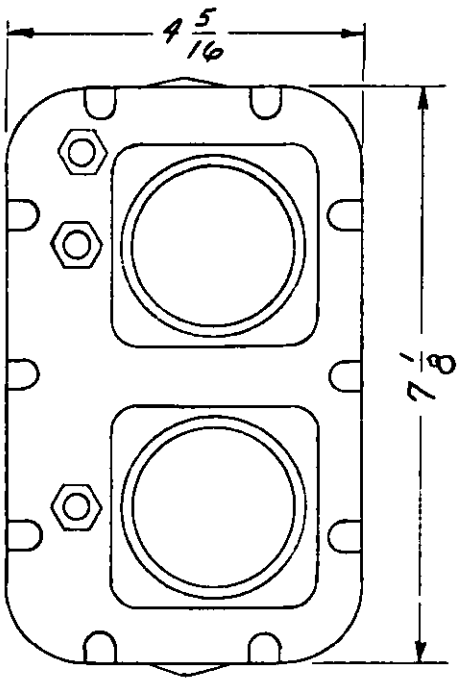


FIGURE 1
OUTLINE DRAWING
CU-395/ARN-21A
SCALE 1/2 = 1 (APPROX)

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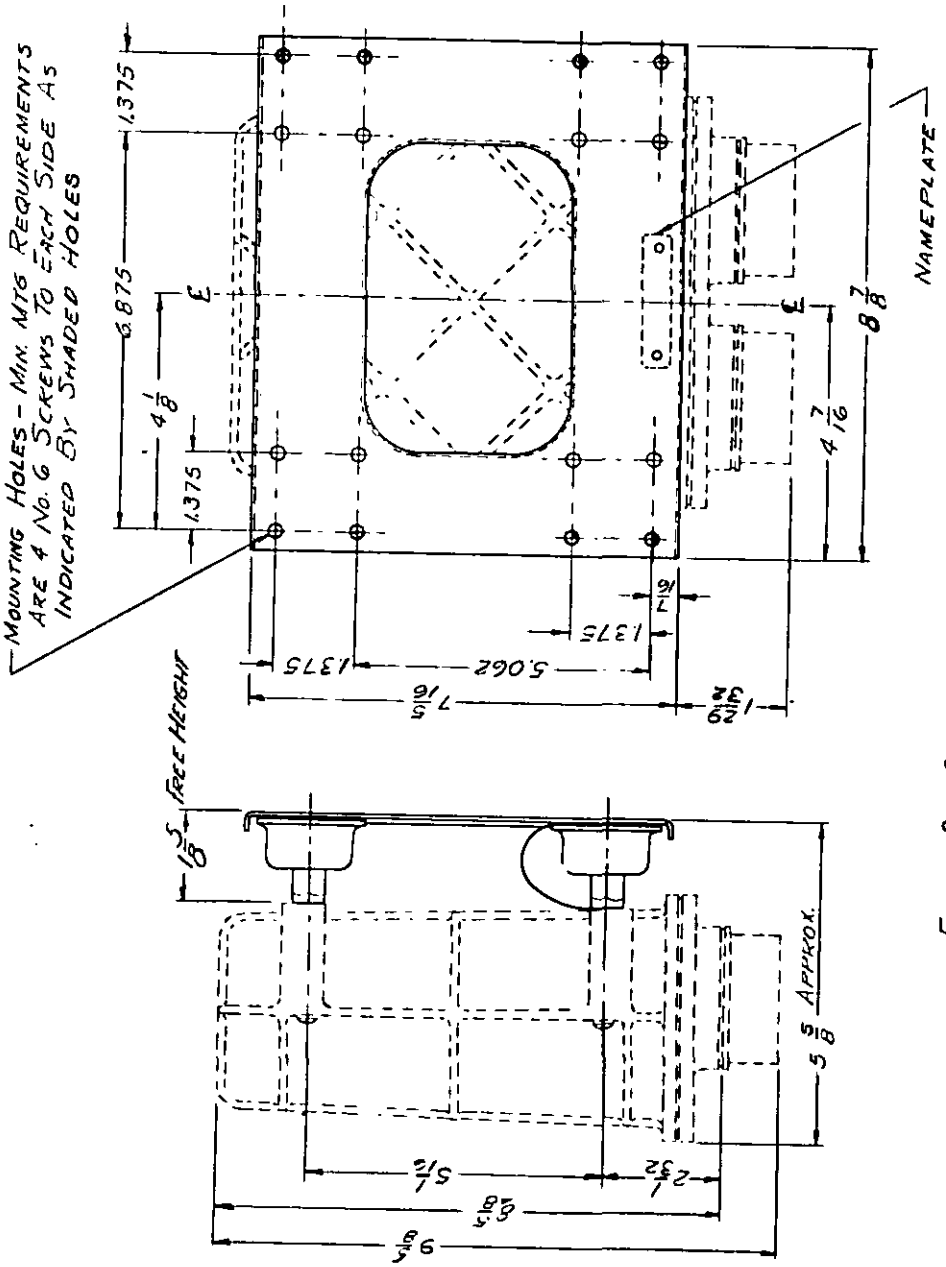


FIGURE 2: OUTLINE DRAWING MOUNTING MT-1530/ARN-21A SCALE: 1/2 = 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD. FORM 1426 REPLACES NAVSHIPS FORM 4863 WHICH IS OBSOLETE C-9279

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